Lecture plan · for the lecturer

Integer programming and branch and bound

Optimization · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Model with integers.
- Use relaxations as bounds.
- Run branch and bound.

Activity: Grow the tree

1. Warm-up (5 min): Show 2.5 buses and ask students to vote: round up, round down, or something else.
2. Relax (10 min): Solve the knapsack relaxation by value-to-weight ratio.
3. Branch (25 min): Pairs draw the tree, computing bounds and pruning nodes.
4. Compare (15 min): Compare the rounded answer, the proven optimum and the gap.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- How many nodes did you prune?
- How far was rounding from optimal?
- What does the gap tell a manager?

Exit ticket

- Name one decision in your institution that needs integer variables.

Try it in your next lecture
Give pairs a four-item knapsack and ask them to round the relaxation first; then draw the branch-and-bound tree together and compare the rounded answer with the proven optimum.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Maximisation: best whole solution so far 13; a branch's relaxation gives 12. What next?**

◯ Branch further ◯ Prune it ◯ Round it

2 **Solve the knapsack.**

Items (weight, value): a (5, 10), b (4, 7), c (3, 6); capacity 7. The LP relaxation gives _____ and the integer optimum is _____.

Word bank: 10 · 13 · 14 · 16

3 **Which statements about branch and bound are true? Choose all that apply.**

☐ It branches on a fractional variable ☐ It prunes nodes whose bound cannot beat the incumbent

☐ The bound–incumbent gap certifies quality ☐ It always enumerates every combination

4 **Rounding down an optimal LP relaxation can give an integer solution far from the integer optimum.**

◯ True ◯ False

5 **Order what branch and bound does at one node.**

☐ If its bound cannot beat the incumbent, prune the node

☐ Solve the node's LP relaxation

☐ Otherwise, branch on a fractional variable

☐ Otherwise, if its solution is integral, make it the new incumbent

Answers · for the lecturer

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1 **Prune it**

In a maximisation, 12 is an upper bound for the whole branch, so nothing in it can beat 13.

2 **Items (weight, value): a (5, 10), b (4, 7), c (3, 6); capacity 7. The LP relaxation gives 14 and the integer optimum is 13.**

Relaxation: a plus 2/3 of c = 14. Integer: b + c = 13; a alone gives 10.

3 **It branches on a fractional variable; It prunes nodes whose bound cannot beat the incumbent; The bound–incumbent gap certifies quality**

Pruning is exactly what avoids full enumeration.

4 **True**

In the example, rounding gives 10 while the optimum is 13.

5 **Top to bottom: 2, 1, 4, 3**

Bound first: a node that cannot beat the incumbent is pruned at once; an integral solution becomes the new incumbent and needs no branching; only a fractional one is split.

For you to think about

Do your students ever see a rounded solution fail? Which example from your context would show it?